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DXARTS 430
Prof. Castillo
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Green Tension

Videos/Photo:

<https://drive.google.com/drive/folders/17FMNF-Adh9z3zsLmzytZfjx8L34leCka?usp=sharing>

Github (code): <https://github.com/RemiVernon/Green-Tension>

Miro: https://miro.com/app/board/uXjVHYKxPoY=?share_link_id=8988375827

Core Narrative

- The core narrative of green tension is to use algorithms to demonstrate how human intervention with natural ecosystems harms the progression and health of those systems.
- By taking data of human activity relative to the plant, the plant movement is impacted by the human data.
- This is inspired by how humans create things like industrial facilities in spaces where natural ecosystems may have existed previously. Human activity destroys natural environments.
- The goal of the project is to imitate this interaction between classes and use algorithms to create a visual display of that unbalanced relationship.

Use of algorithms:

- The use of an L-system creates a tree graphic that is based on a series of growth rules. The tree grows based on the rules of the L-system formula and the data that it receives from the ultrasonic sensor.
- The ultrasonic sensor sends data readings based on how close humans are to the sensor. The stepper motor pulley system is triggered by the data readings that come from the sensor. This means that the plant is being pulled based on the numbers being received by the sensor.
- When a human is close to the ultrasonic sensor, the digital graphic of the tree from p5js will also turn red and delete quickly. When humans are far from the sensor, the tree will slowly grow and turn green.
- Moreover, the ultrasonic sensor triggers audio frequencies that play from a speaker. The frequencies and timing of the beeps are connected to the distance reading from the ultrasonic sensor. Frequencies are randomized and beep faster when humans are close to the sensor, but when humans are far from the ultrasonic sensor, the beeping is slow and at a low tone frequency.

Software/Hardware:

- The ultrasonic sensor is connected to an arduino that is connected to a stepper motor, speaker, and p5js. All three variables are controlled by the ultrasonic sensor. The p5js and arduino connection is built in so that whenever the p5js code runs, the preview will ask for a confirming click to allow the arduino to control the graphic, or send its data.

Aesthetic:

- The aesthetic of the project draws from 80s video game visuals that consists of blocky, digitalized, depictions of real life objects.
- The use of an old TV is meant to symbolize how this interaction between humans and natural ecosystems is not a new phenomenon and that this has been happening throughout human history.
- The pulley system to the left of the plant is meant to resemble an industrial plant while the ultrasonic sensor resembles the eyes of a living organism. The TV on the right represents the digital perception of the wilting tree.

Presentation:

- The project was presented as an installation in a white wall gallery space.
- The plant, pulley system, and tv screen display were placed on pedestals.
- Audiences naturally interacted with the piece and learned about their relationship with it as they spent more time with the system. This was done through interactive indicators like the graphic, the moving plant, and the beeping frequencies.